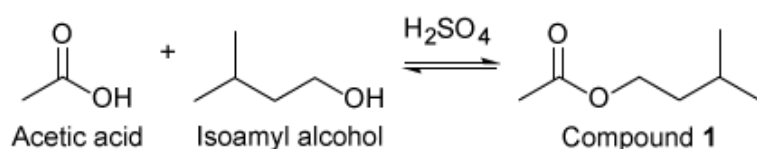


Esters are carboxylic acid derivatives commonly found in fragrances and artificial flavorings and are responsible for the flavor and scent in fruits. Esters are prepared through a condensation reaction between a carboxylic acid and an alcohol. An example is shown in Reaction 1.



Reaction 1

Table 1 shows the scent and boiling point of esters resulting from different carboxylic acid and alcohol condensations.

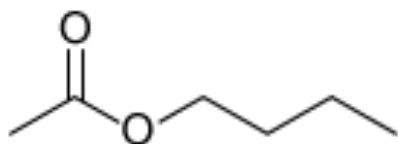
Table 1 Ester Components and Characteristics

Carboxylic acid	Alcohol		Ester	Scent	Ester boiling point
Acetic acid	Isoamyl alcohol	→	Compound 1	Banana	142 °C
Acetic acid	Propanol	→	Compound 2	Pear	102 °C
Butanoic acid	Ethanol	→	Compound 3	Pineapple	121 °C

Students synthesized an ester using one of the carboxylic acid/alcohol combinations shown in Table 1. They added 40 mmol carboxylic acid and 15 mmol alcohol to a 25-mL round-bottom flask. While stirring, 0.30 mL concentrated sulfuric acid was added dropwise. The reaction was then heated to reflux for an hour. After cooling the reaction mixture, the reaction product was isolated via extraction and purified by simple distillation. The reaction product was verified by electron-impact mass spectrometry, which ionizes the molecule by removing an electron.

The structure of an ester is shown below. Does this structure correspond to the ester that has a pineapple scent as described in

Table 1?



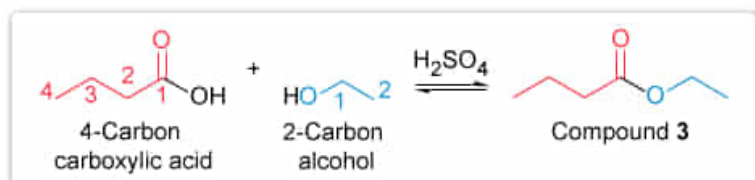
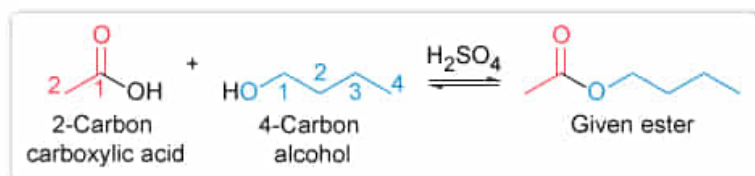
- ☐ A. Yes, because this is the structure of Compound **3**, which is made from a four-carbon carboxylic acid and a two-carbon alcohol. [13%]
- ✓ ☒ B. No, because this ester was made from a two-carbon carboxylic acid and a four-carbon alcohol and does not correspond to Compound **3**. [70%]
- ☐ C. Yes, because this is the structure of Compound **3**, which is made from a two-carbon carboxylic acid and a four-carbon alcohol. [6%]
- ☐ D. No, because this is the structure of Compound **2**, which has a pear scent. [10%]

Correct

69% answered correctly

Explanation:

Ester formation



The given ester and Compound **3** are made from different carboxylic acids and alcohols. Therefore, the given ester does *not* match the structure of Compound **3**.

Esters are **carboxylic acid derivatives** made from a carboxylic acid and an alcohol via a **condensation reaction**. **Fischer esterification** is an acid-catalyzed esterification reaction where the acid catalyst protonates the carbonyl oxygen on the carboxylic acid, making the carbonyl more electrophilic and susceptible to nucleophilic attack. The alcohol forms a bond with the carbonyl of the carboxylic acid via nucleophilic addition. The carboxylic acid -OH group is then protonated, creating a good leaving group (H_2O). Finally, an ester is formed upon H_2O elimination. As such, the **carbon chain** on the ester that **includes the carbonyl** carbon stems from the **carboxylic acid**, and the **carbon chain bonded to the oxygen stems from the alcohol**.

Based on Table 1, Compound **3** is the ester with the pineapple scent and is made from butanoic acid and ethanol. The **prefix *but-*** indicates the carboxylic acid has four carbons and the prefix ***eth-*** indicates the alcohol has two carbons. Therefore, when butanoic acid and ethanol react, the portion of the resulting ester containing the carbonyl has four carbons and the carbon chain bonded to the oxygen atom has two carbons. The ester given in this question shows the reverse: the portion of the ester containing the carbonyl has *two* carbons and the carbon chain bonded to the oxygen atom has *four* carbons.

Therefore, **the given ester is not Compound 3**, the ester with a pineapple scent.

(Choice A) While Compound **3** is the ester that has a pineapple scent and is made from a four-carbon carboxylic acid and a two-carbon alcohol, the given structure does *not* correspond to Compound **3**.

(Choice C) While the given ester is made from a two-carbon carboxylic acid and a four-carbon alcohol, its structure does *not*

correspond to the ester with the pineapple scent. The number of carbons in the carboxylic acid and alcohol have been reversed.

(Choice D) According to Table 1, Compound **2** is made from acetic acid (a two-carbon carboxylic acid) and propanol (a three-carbon alcohol). However, the given structure is made from a *four*-carbon alcohol.

Educational objective:

Esters are carboxylic acid derivatives made from a carboxylic acid and alcohol via a condensation reaction, including a Fischer esterification. The carbon chain on an ester that includes the carbonyl stems from the carboxylic acid, and the carbon chain bonded to the oxygen comes from the alcohol.

Time spent:0 sec

Subject:

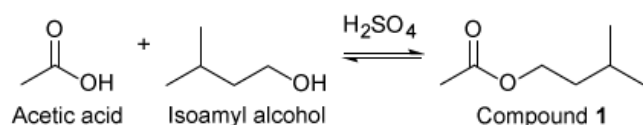
Organic Chemistry

QID:403926

System:

Functional Groups and Their Reactions

Esters are carboxylic acid derivatives commonly found in fragrances and artificial flavorings and are responsible for the flavor and scent in fruits. Esters are prepared through a condensation reaction between a carboxylic acid and an alcohol. An example is shown in Reaction 1.



Reaction 1

Table 1 shows the scent and boiling point of esters resulting from different carboxylic acid and alcohol condensations.

Table 1 Ester Components and Characteristics

Carboxylic acid	Alcohol		Ester	Scent	Ester boiling point
Acetic acid	Isoamyl alcohol	→	Compound 1	Banana	142 °C
Acetic acid	Propanol	→	Compound 2	Pear	102 °C
Butanoic acid	Ethanol	→	Compound 3	Pineapple	121 °C

Students synthesized an ester using one of the carboxylic acid/alcohol combinations shown in Table 1. They added 40 mmol carboxylic acid and 15 mmol alcohol to a 25-mL round-bottom flask. While stirring, 0.30 mL concentrated sulfuric acid was added dropwise. The reaction was then heated to reflux for an hour. After cooling the reaction mixture, the reaction product was isolated via extraction and purified by simple distillation. The reaction product was verified by electron-impact mass spectrometry, which ionizes the molecule by removing an electron.

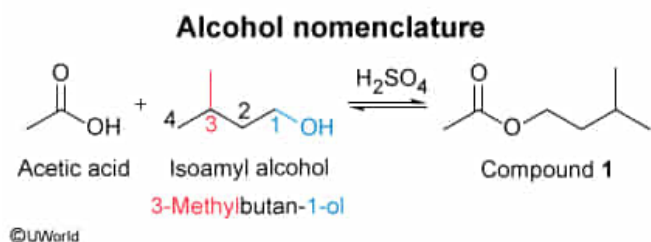
What is the IUPAC name of isoamyl alcohol based on the structure shown in Reaction 1?

- ✓ ☒ A. 3-Methylbutan-1-ol [79%]
☐ B. 2-Methylbutan-4-ol [11%]
☐ C. 2-Isopropylethan-1-ol [4%]
☐ D. 3,3-Dimethylpropan-1-ol [4%]

Correct

80% answered correctly

Explanation:



Alcohols are molecules containing a **hydroxyl group** ($-\text{OH}$) that are named using systematic rules from the IUPAC. For molecules where $-\text{OH}$ is the **highest priority functional group**, the longest continuous carbon chain containing the hydroxyl group is the parent chain. The **prefix** in the molecule's name corresponds to the number of carbon atoms in the parent chain, and the **suffix** $-ol$ corresponds to the alcohol functional group. The parent chain is numbered so that the hydroxyl group is given the lowest possible position, and the location of the hydroxyl group is indicated right before the suffix. Substituents are listed before the parent chain name using a number to indicate its position on the parent chain.

Reaction 1 shows the Fischer esterification between acetic acid and isoamyl alcohol. Based on the structure of isoamyl alcohol given in the reaction scheme, the parent chain contains four carbons (prefix *but-*) and is numbered so the hydroxyl group is given the lowest position number possible (carbon 1). There is a methyl substituent on carbon 3. Therefore, the IUPAC name of isoamyl alcohol is **3-methylbutan-1-ol**.

(Choice B) 2-Methylbutan-4-ol **incorrectly numbers** the parent chain by giving the hydroxyl group the *highest* instead of lowest possible position number.

(Choice C) 2-Isopropylethan-1-ol **incorrectly identifies** the parent chain as two carbons long and the substituent as an **isopropyl group**. The longest continuous carbon chain is *four* carbons long and the substituent is a *methyl* group.

(Choice D) 3,3-Dimethylpropan-1-ol **incorrectly identifies** the parent chain as three carbons long and includes an additional methyl substituent. The longest continuous carbon chain is *four* carbons.

Educational objective:

Alcohols are molecules containing a hydroxyl group (-OH) and are named using the suffix -ol . The prefix indicates the length of the parent chain (longest continuous carbon chain containing the hydroxyl group), and the position of the hydroxyl group is indicated by its position number before the suffix.

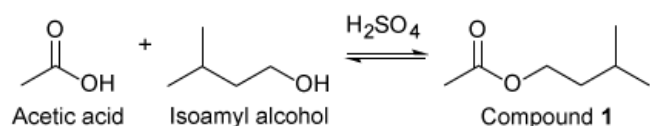
Time spent:0 sec

Subject:
Organic Chemistry

QID:403927

System:
Functional Groups and Their Reactions

Esters are carboxylic acid derivatives commonly found in fragrances and artificial flavorings and are responsible for the flavor and scent in fruits. Esters are prepared through a condensation reaction between a carboxylic acid and an alcohol. An example is shown in Reaction 1.



Reaction 1

Table 1 shows the scent and boiling point of esters resulting from different carboxylic acid and alcohol condensations.

Table 1 Ester Components and Characteristics

Carboxylic acid	Alcohol		Ester	Scent	Ester boiling point
Acetic acid	Isoamyl alcohol	→	Compound 1	Banana	142 °C
Acetic acid	Propanol	→	Compound 2	Pear	102 °C
Butanoic acid	Ethanol	→	Compound 3	Pineapple	121 °C

Students synthesized an ester using one of the carboxylic acid/alcohol combinations shown in Table 1. They added 40 mmol carboxylic acid and 15 mmol alcohol to a 25-mL round-bottom flask. While stirring, 0.30 mL concentrated sulfuric acid was added dropwise. The reaction was then heated to reflux for an hour. After cooling the reaction mixture, the reaction product was isolated via extraction and purified by simple distillation. The reaction product was verified by electron-impact mass spectrometry, which ionizes the molecule by removing an electron.

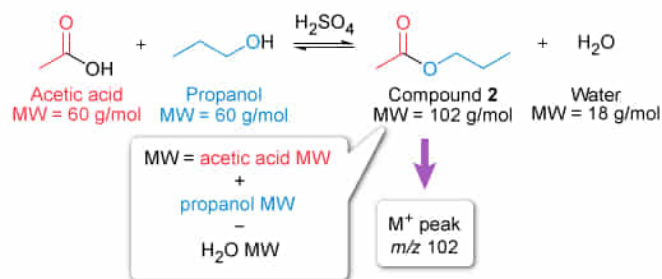
When a student uses mass spectrometry to verify that they made Compound 2, what is the expected m/z of the molecular ion peak (M^+) for Compound 2? (Note: Acetic acid and propanol each have a molecular weight of 60 g/mol.)

- ☐ A. 60 [13%]
☒ B. 102 [40%]
☐ C. 120 [43%]
☐ D. 138 [2%]

Explanation:

Fischer esterification and the molecular ion (M^+) peak

- Fischer esterification joins together a **carboxylic acid** and an **alcohol**, resulting in the loss of H_2O .
- The M^+ peak has a m/z equal to the molecular weight (MW) of the analyzed compound.



© UWorld

A **Fischer esterification** is a **condensation reaction** between a **carboxylic acid** and an **alcohol** that produces an ester (a **carboxylic acid derivative**) and results in the **loss of water**. Different lab techniques can be used to identify whether the desired reaction product was produced. For example, **mass spectrometry** analyzes the mass of molecules and molecular fragments and is a useful technique if the reaction product has a different molecular weight than the starting materials or has characteristic fragmentation patterns.

The passage states that the students verified the reaction product by electron-impact mass spectrometry, in which the molecule is ionized by the removal of an electron. The resulting ion is the **molecular ion (M^+)** and is a radical cation with a **mass-to-charge ratio (m/z)** equal to the molecular weight of the molecule because the mass of the electron lost is negligible.

Table 1 shows that Compound 2 was prepared by a condensation reaction between acetic acid and propanol. Because condensation reactions join two compounds together, resulting in the loss of water, the molecular weight of Compound 2 is obtained by adding together the molecular weights of acetic acid and propanol given in the question and subtracting the molecular weight of water (18 g/mol):

$$\text{Compound 2 molecular weight} = 60 \text{ g/mol} + 60 \text{ g/mol} - 18 \text{ g/mol} = 102 \text{ g/mol}$$

Therefore, the expected m/z for the M^+ for Compound 2 is 102.

(Choice A) Because acetic acid and propanol have molecular weights of 60 g/mol, the M^+ for each of these molecules has m/z 60. Observing an M^+ with m/z 60 indicates that Compound **2** did *not* form.

(Choice C) A m/z value of 120 results from adding the molecular weights of acetic acid and propanol. Because the condensation between acetic acid and propanol results in *loss of water*, the M^+ m/z for Compound **2** must be *less* than 120.

(Choice D) A m/z value of 138 results from incorrectly *adding* the mass of water to the sum of the molecular weights to acetic acid and propanol.

Educational objective:

Esters are made by the condensation of a carboxylic acid and an alcohol, resulting in the loss of water. The molecular weight of ester can be calculated by adding together the molecular weights of its constituent carboxylic acid and alcohol and subtracting the molecular weight of water.

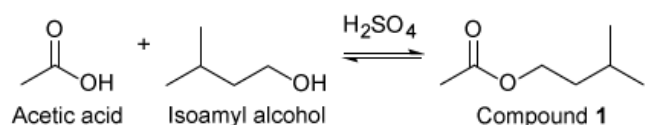
Time spent:0 sec

Subject:
Organic Chemistry

QID:403928

System:
Functional Groups and Their Reactions

Esters are carboxylic acid derivatives commonly found in fragrances and artificial flavorings and are responsible for the flavor and scent in fruits. Esters are prepared through a condensation reaction between a carboxylic acid and an alcohol. An example is shown in Reaction 1.



Reaction 1

Table 1 shows the scent and boiling point of esters resulting from different carboxylic acid and alcohol condensations.

Table 1 Ester Components and Characteristics

Carboxylic acid	Alcohol		Ester	Scent	Ester boiling point
Acetic acid	Isoamyl alcohol	→	Compound 1	Banana	142 °C
Acetic acid	Propanol	→	Compound 2	Pear	102 °C
Butanoic acid	Ethanol	→	Compound 3	Pineapple	121 °C

Students synthesized an ester using one of the carboxylic acid/alcohol combinations shown in Table 1. They added 40 mmol carboxylic acid and 15 mmol alcohol to a 25-mL round-bottom flask. While stirring, 0.30 mL concentrated sulfuric acid was added dropwise. The reaction was then heated to reflux for an hour. After cooling the reaction mixture, the reaction product was isolated via extraction and purified by simple distillation. The reaction product was verified by electron-impact mass spectrometry, which ionizes the molecule by removing an electron.

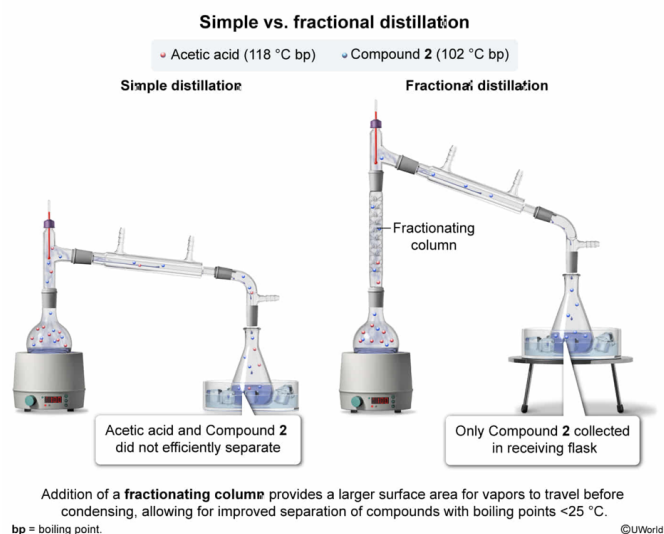
A student purifies Compound **2** but discovers that a simple distillation was not successful in separating the mixture of Compound **2** and acetic acid (boiling point = 118 °C). Which of the following changes to the simple distillation setup would allow for separation of Compound **2** and acetic acid?

- ✔ ☒ A. Add a fractionating column [69%]
- ☐ B. Place the receiving flask in an ice bath [9%]
- ☐ C. Lower the position of the thermometer in the distillation setup [2%]
- ☐ D. Connect the distillation setup to a vacuum [19%]

Correct

68% answered correctly

Explanation:



Separating a mixture requires a technique suitable to the properties (eg, boiling point, polarity) of the mixture components. **Distillation** is used to separate a mixture of liquids based on the boiling point of the mixture components. The mixture is heated slowly and the molecule with the lower boiling point evaporates first. As the vapors travel toward the thermometer and condenser, the vapors condense and are collected in the receiving flask as distillate. There are three common types of distillation: **simple**, **fractional**, and **vacuum**.

The passage states that the esters were purified by simple distillation. Table 1 shows the boiling point of Compound 2 is 102 °C, and the question states the boiling point of acetic acid is 118 °C. The difference between the boiling points is

$$\text{Boiling point difference} = 118\text{ °C} - 102\text{ °C} = 16\text{ °C}$$

Fractional distillation, which is ideal for separating compounds with a **boiling point difference <25 °C**, is the best method to separate Compound 2 and acetic acid. Therefore, **adding a fractionating column** to the distillation setup changes the simple distillation to a fractional distillation. The fractionating column is added between the flask and condenser to provide a larger surface area for the distillate vapors to travel before collection in the receiving flask.

(Choice B) Placing the receiving flask in an ice bath keeps the distillate from evaporating. The ice bath does not affect the separation of the mixture components.

(Choice C) The thermometer is placed to give the temperature of the vapors before entering the condenser. Lowering the position of the thermometer in the distillation setup will not affect the distillation process itself and therefore will *not* improve separation of the mixture.

(Choice D) Connecting the distillation setup to a vacuum changes the distillation from a simple to a vacuum distillation. While a vacuum reduces the pressure of the system and lowers the boiling point of the mixture components, vacuum distillation does not generally improve the separation efficiency of a mixture with boiling points $<25\text{ }^{\circ}\text{C}$.

Educational objective:

Separating a mixture requires the use of a technique suitable to the properties of the mixture components. Distillation is used to separate a mixture of liquids based on boiling point. Fractional distillation is ideal for liquids with a boiling point difference $<25\text{ }^{\circ}\text{C}$.

Time spent:0 sec

Subject:
Organic Chemistry

QID:403929

System:
Functional Groups and Their Reactions